

Cool People — My Next Best Health Action

Team: Cool People · **Environment:** HUM-HAC-113 · **Problem Statement:** STARS: My Next Best Health Action

1. One-liner

A member-facing agent that doesn't list your care gaps — it **decides which one matters most for you right now, says why it rejected the others, and books it**. Coverage decisions stay with a licensed human.

The sentence we want a leader to repeat:

"The agent knows which gap to push, and knows which questions it isn't allowed to answer."

2. The problem

Members with open care gaps get generic, blanket outreach. Four gaps → four mailers → zero closes. Nobody ranks. The plan wastes spend on low-weight measures and on channels the member has already ignored; the member gets noise instead of a next step.

3. Name the baseline (say this out loud — in both rounds)

"The obvious build here is a chatbot that lists your gaps. We think that's the thing that already doesn't work — a mailer is a list. So we built the ranking layer instead."

Why this line exists: in Round 1 the contrast is visible (four other teams share this problem statement). In Round 2 it is not — the other nine finalists are on different problem statements, and TLT never saw the chatbots we beat. Without naming the baseline, our differentiator is legible only to us. Five seconds, both rounds, non-negotiable.

4. The solution: ranking, not retrieval

“Next Best” is the entire ask. The agent scores each open gap on:

- **Star measure weight** — 1x vs 3x; 3x measures move plan revenue
- **Days to close** — urgency
- **Member/segment response history** — which channels actually worked

...and surfaces **one** recommendation with a stated reason and an explicit `rejected_because` for the rest.

5. Architecture (Google ADK)

Orchestrator (root) — routes to a sub-agent, and enforces the compliance gate.

Prioritizer Agent

The differentiator. Also a reusable next-best-action ranking service.

Tool	Reads	Returns
<code>get_open_gaps</code>	<code>care_gaps</code>	open gaps + days-to-close
<code>get_measure_weight</code>	<code>stars_performance</code>	Star weight per measure
<code>get_response_history</code>	<code>historical_interventions</code> , <code>campaign_dispositions</code>	channels this member/segment responded to

Actioner Agent

One agent, three tools — depth without three integration surfaces.

Tool	Reads	Returns
<code>explain_gap</code>	<code>coverage_rules</code> , call transcripts	why it matters, what it costs
<code>find_provider</code>	<code>providers</code> , <code>members</code>	nearest in-network provider
<code>book_or_callback</code>	<code>appointment_slots</code>	simulated confirmation

Compliance gate (orchestrator level)

Any coverage **determination** request → agent states the general rule from `coverage_rules` , refuses the determination, routes to a human advocate, logs to `compliance_flags` .

6. Score decomposition — the top-5 move

The agent must not merely *assert* its pick. It **shows the arithmetic on screen**:

```
Diabetic Eye Exam    weight 3.0 × urgency 0.85 × propensity 0.72 = 1.84    ← SEL
Flu Shot             weight 1.0 × urgency 0.40 × propensity 0.15 = 0.06
CBP Follow-up        weight 3.0 × urgency 0.20 × propensity 0.30 = 0.18
```

Why this is the highest-leverage 20 minutes left in the build: it converts a chat flourish into an **auditable decision system**. A judge no longer takes our word for it. This is the difference between a demo and something Humana could actually run — and it speaks directly to the Round 2 rubric’s demand for grounded, controlled AI.

7. Rubric-legible naming (deck language, not code changes)

We already built four of the things the Round 2 “Initial AI Approach” rubric explicitly rewards. We were not calling them by name.

What we built	What we call it
Scoring formula	Risk / propensity scoring
Orchestrator selecting a sub-agent	Agent routing
Compliance gate	Classification + confidence-gated escalation
<code>reason</code> / <code>rejected_because</code> fields	Structured outputs
<code>explain_gap</code> over transcripts + <code>coverage_rules</code>	Grounded retrieval

Same build. Rubric-legible. Zero additional build cost.

8. Business impact — compute from real data, do not ship a vibe

Target sentence: “N members carry open [3x-weighted measure] gaps. Closing k% moves the measure from X.X to Y.Y stars. Per-member targeted outreach replaces blanket mailing across the full segment.”

Source: `stars_performance + segment_performance + care_gaps` **Owner assigned by 10:00. One real number, one slide.** Non-negotiable — this is the Round 2 rubric’s top line, and Rating 5 requires a credible path to *measurable* impact.

9. Reusability (Round 2 rubric)

The Prioritizer is **not** a care-gap chatbot. It is a **next-best-action ranking service**: swap the scoring inputs and it ranks interventions for advocates, or actions for providers. The compliance gate is a **reusable enterprise control** for any member-facing AI at Humana.

Zero additional build cost. This is framing, and it moves Feasibility / Reusability / Scalability from 3 to 4–5.

10. Scope: explicitly out

- Real scheduling integration → simulated
- Write-back to campaign systems → that’s the adjacent problem statement
- Voice → web chat only
- Any coverage determination → **by design**

Say this in Q&A. Stated tradeoffs read as maturity.

11. Fallback — decide by 9:45, not noon

If `historical_interventions / segment_performance` are thin, scoring degrades to:

Star weight × days-to-close × gap age

The rejection moment survives. We lose “you responded to SMS,” we keep “not your flu shot — it’s single-weighted.” The score decomposition still renders. **Profile first, commit**

early.

12. Timeline — Day 1 (recording due 4:30)

Time	Milestone
9:00– 9:20	Env up, unzip unstructured data
9:20– 9:45	Profile all tables. Row counts, join keys, does per-member response history exist? → fallback decision
9:45– 10:15	Lock scope. Pick the one hero member whose data makes the rejection land. Write the demo script <i>now</i>
10:00 →	Impact-number owner starts in parallel
10:15– 12:00	Prioritizer + BigQuery tools. Our moat. Working first
12:00– 1:00	Lunch / transcript objection mining
1:00– 2:30	Actioner (3 tools) + orchestrator + compliance gate
2:30– 2:50	Score decomposition UI
2:50– 3:30	Freeze. Polish, dry-run 3×
3:30– 4:30	Record. Buffer for the recording failing (it will)

13. Day 2 — Round 1 (10:00 · 5 min + 3 min Q&A)

Scored on: Problem Understanding · Technical Ambition · Creativity · Presentation

Time	Beat
	Name the baseline — “the obvious build is a chatbot that lists gaps; a mailer is a

0:20	list"
0:25	Problem: 4 gaps, 4 mailers, 0 closes
0:30	Ask: "What should I do about my health?"
1:00	The moment — agent rejects out loud, decomposition visible : "Not your flu shot — single-weighted, you ignored two mailers. Your diabetic eye exam: triple-weighted, 40 days to close, you booked by SMS last October."
1:00	Action — explains plainly, finds provider 3.2 mi in-network, books Tuesday 10am
0:45	Guardrail — "Is this covered 100%?" → rule, refusal, route to advocate
0:40	Impact number

14. Day 2 — Round 2 (2:00 · 6 min + 2 min Q&A · one intern presents to TLT)

Different rubric: Solution Impact · Initial AI Approach · User Experience · Feasibility / Reusability / Scalability. Creativity and Technical Ambition are no longer scored. Rebalance accordingly.

Time	Beat
0:20	Name the baseline — TLT did not see the four chatbots we beat
1:20	Lead with impact. The number, first
1:00	Governance as a design principle — the agent knows what it must not decide. Best asset with this audience
0:50	The rejection moment, compressed — decomposition on screen
0:50	Action flow
0:40	AI approach, named in rubric language: routing, propensity scoring, grounded retrieval, confidence-gated escalation, structured outputs
1:00	Reusability: NBA ranking service + reusable compliance control → enterprise path

Presenter: whoever tells the **impact** story best. Not whoever wrote the most code.

15. Play the recording

Day 2 requires a live **presentation**, not a live **demo**. Live LLM agents fail in front of judges — latency, cold start, a bad sample. Narrate over the recorded perfect run.

If a judge asks *“is it real?”* — run it live in Q&A, where failure costs nothing.

16. Language discipline

- Say: *“Coverage determinations stay with a licensed human — the agent explains and routes, it never decides.”*
- **Do not** name specific litigation in the room. Same point, without sounding like we’re litigating our employer.